جامعة الشارقة
UNIVERSITY OF SHARJAH

Research Fund Application Form

Office of Vice Chancellor for Research and Graduate Studies

Funding Institute:

Research Institute of Medical and Health Sciences	☐
Research Institute of Sciences and Engineering	☒
Research Institute of Humanities and Social Sciences	☐

Project ID Number												

Project Title: Leveraging Knowledge Graphs and Generative AI for Augmented Research Paper Retrieval
عنوان البحث باللغة العربية

Type of Fund (Select one)

☒ Seed Research Grant (New faculty)

☐ Competitive Research Grant

☐ Collaborative Research Grant with Public & Private Organizations

- Targeted Research Grant fo

Researchers Participation:

☒ One Department ☐ More than One Department in One College ☐ Several Colleges in UoS ☐ External Participants

Principal Investigator (PI) (PI may not hold UOS funding for more than two projects at the same time)

Full Name: Danilo Dessi Signature: Danilo Dessi University ID: 102191

Academic Rank: Assistant Prof ☒ Associate Prof ☐ Professor ☐

College: College of Computing and Informatics Department: Computer Science

Telephone 3537 Fax : Email: ddessi@sharjah.ac.ae

On-Going Research Projects Involving the PI (Include Internal and Externally Funded projects)

No.	Project Title / Type	Source of Funds	Amount of Funds	Period	
				From	To
1					
2					
3					

Internal Co-Investigator(s)

The project does not have internal co-investigators. However, given the overlapping research interests of colleagues at the College of Computing and Informatics, there is a good probability that colleagues might in future participate and contribute to the scheduled activities.

External Co-Investigator(s)(For collaborative Research Projects: Please attach an official letter confirming the amount of the funds contributed by the external organization to the project and authorizing the nominated external individuals to participate in the project)

The project does not have formal external collaborators.

1. Abstract

Present here a brief abstract of your research project (Maximum 250 words)

a. Abstract in English

The rapid growth of scientific literature poses an increasing challenge for researchers to discover, access, and assimilate relevant knowledge effectively. While existing scholarly tools provide basic search functionalities, they often fail to recommend contextually relevant, high-quality research papers. Generative models, though promising, have proven unsuitable for working with scholarly knowledge due to issues such as hallucination, which results in the generation of non-factual information. Knowledge Graphs (KGs), which integrate structured data from large-scale scholarly resources, offer new possibilities to enhance search relevance, identify trends, and improve content comprehension. Notable examples include the Computer Science Knowledge Graph (CS-KG) and the Open Research Knowledge Graph (ORKG). This project proposes leveraging these KGs alongside advanced artificial intelligence techniques such as Retrieval-Augmented Generation (RAG) to develop enhanced tools for researchers. The focus will be on improving the retrieval of relevant publications based on abstracts and titles, enabling more precise and comprehensive research recommendations. By harnessing the knowledge embedded in these graphs, the project aims to create novel tools that assist researchers in navigating the ever-growing body of scientific literature, bridging the gap between static search tools and dynamic, knowledge-aware research assistants.

b. Abstract in Arabic

النمو السريع للأدبيات العلمية يفرض تحديًا متزايدًا على الباحثين لاكتشاف والوصول واستيعاب المعرفة ذات الصلة بشكل فعال. على الرغم من أن الأدوات العلمية الحالية توفر وظائف بحثية أساسية، إلا أنها غالبًا ما تفشل في تقديم توصيات لأبحاث ذات جودة عالية وتتناسب مع السياق. بينما تعد النماذج التوليدية واعدة، فقد ثبت عدم ملاءمتها للتعامل مع المعرفة العلمية بسبب مشكلات مثل الهلوسة"، التي تؤدي إلى توليد معلومات غير دقيقة. توفر الرسوم البيانية المعرفية" (KGs)، التي تدمج بيانات منظمة من موارد علمية واسعة النطاق، إمكانيات جديدة لتعزيز دقة البحث، وتحديد الاتجاهات، وتحسين فهم المحتوى. تشمل الأمثلة البارزة على ذلك الرسم البياني المعرفي لعلوم الحاسوب (CS-KG) والرسم البياني للمعرفة البحثية المفتوحة (ORKG). يقترح هذا المشروع لتطوير (RAG) "الاستفادة من هذه الرسوم البيانية إلى جانب تقنيات الذكاء الاصطناعي المتقدمة مثل "التوليد المدعوم بالاسترجاع أدوات محسنة للباحثين. سيركز المشروع على تحسين استرجاع المنشورات ذات الصلة بناءً على الملخصات والعناوين، مما يتيح توصيات بحثية أكثر دقة وشمولية. من خلال استغلال المعرفة المضمنة في هذه الرسوم البيانية، يهدف المشروع إلى إنشاء أدوات جديدة تساعد الباحثين في التنقل عبر الكم المتزايد من الأدبيات العلمية، وبالتالي سد الفجوة بين أدوات البحث الثابتة والمساعدات البحثية الديناميكية المدركة للمعرفة.

2. Problem formulation and Project Objectives

a. Problem formulation

Given a query *q*, which can consist of a paper title or paper title+abstract, the objective is to retrieve a set of relevant publications $P=\{p_1, p_2, \ldots, p_n\}$ where each p_i is associated with a score indicating its relevance to q. Traditional keyword-based retrieval methods denoted as R_{kw}, often fail by ignoring the semantic relationships between entities and the broader context of the query. Let $KG=(E, R)$ represent a Knowledge Graph, where *E* is the set of entities (e.g., tasks, methods, materials, metrics) and *V* is the set of semantic relationships connecting these entities. In this project, *K*G refers specifically to the Computer Science Knowledge Graph (CS-KG)[1], which contains structured information extracted from research article titles and abstracts. We aim to integrate a generative model *M*, specifically a Retrieval-Augmented Generation (RAG) model, with the knowledge graph *KG* to create a system that generates contextually relevant search results based on q. The retrieval process denoted as $RKG(q, G, M)$, combines the structured knowledge of *KG* and the generative capabilities of *M* to enhance the search relevance. The system's goal is to return a ranked set of publications P by leveraging both the semantic structure of *KG* and the contextual understanding provided by *M*. Formally, for each query q, we seek to maximize the relevance score $S(p_i,q)$ *(see eq. 1)* for each publication $p_i \in P$, where the score is a function of both the relationships in *KG* and the language model *M*.

$$\textbf{eq 1}.\ S(pi, q) = fKG(q, pi, KG) + fGen(q, pi, M)$$

b. Project Objectives

Objective 1. Develop a Knowledge Graph-Enhanced Publication Retrieval System. Design and implement a prototype tool that integrates the CS-KG with RAG to retrieve relevant research publications. This system will use structured knowledge to generate search results that are more contextually accurate and relevant to researcher queries based on abstracts and titles.

Objective 2. Improve Search Relevance Through Knowledge Graph and LLM Integration. Leverage the CS-KG to enhance search relevance by connecting research papers through semantic relationships, such as shared topics, data, and methods. This will allow for more meaningful recommendations beyond keyword-based search.

Objective 3. Lay the Foundation for Future Knowledge Graph-Based Research Tools. Document the technical challenges and solutions encountered during the project, creating a roadmap for future research and development of knowledge graph-based scholarly tools. This will include identifying additional functionalities that could further support researchers, such as personalized recommendations or trend analysis.

3. Expected Research Outputs

(Publication of refereed books, book chapters, articles in refereed journals, conferences, or patents, software, products, creative works, etc,…)

- A peer-reviewed journal article detailing the research foundation of the project, including the integration of scholarly knowledge graphs with generative AI models, and the evaluation of the effectiveness and relevance of this integration.

[1] https://w3id.org/cskg/

- A conference/workshop paper within semantic web-related conferences outlining the findings and insights of the integration between knowledge graph technologies and generative AI. Conferences organized within the UAE will be given priority.
- A publicly accessible prototype demonstrating how the tools built on scholarly knowledge graphs and generative AI enhance the retrieval of research publications. The prototype will showcase the practical application of the project's findings.
- A GitHub repository containing all developed source code, datasets, and documentation. This will include a summary of the project's contributions, highlighting the role of the University of Sharjah in advancing scholarly knowledge graph applications and AI-driven research tools.

4. Potential Project Impacts

a. Knowledge Enrichment

This project will significantly contribute to knowledge enrichment in the fields of semantic web technologies, machine learning, and scholarly transformation. By integrating the Computer Science Knowledge Graph (CS-KG) with state-of-the-art generative AI models, the project will produce novel insights into how structured knowledge can enhance scholarly tools. The research will advance the understanding of using knowledge graphs to improve retrieval relevance and contextual understanding in scientific literature. Ultimately, the project seeks to bridge the gap between symbolic and sub-symbolic AI, offering a fresh perspective on their application within the scholarly domain.

b. Importance to Community (Socio-economic impacts)

The project will benefit the broader academic and research community by making it easier for researchers to discover relevant, high-quality literature. This enhanced discovery capability can accelerate research and innovation across disciplines, as researchers spend less time searching for information and more time developing new ideas. On a socio-economic level, this project aligns with the strategic focus on fostering innovation and technological development in higher education and research. By improving access to scholarly knowledge, the project has the potential to raise the quality of research outputs, which in turn can contribute to economic growth by supporting advancements in science, technology, and education. Additionally, by addressing issues such as generative AI model hallucination, the project contributes to refining AI systems to produce more accurate, factual information, ultimately benefiting the general public by reducing the spread of misinformation from state-of-the-art AI tools.

c. Possibility of Innovation and potential commercial product

This project opens possibilities for innovation by creating tools that transform how researchers interact with and discover scholarly information. The integration of CS-KG with generative AI models represents a cutting-edge approach to addressing the growing challenges of literature discovery. The web-based prototype developed through this project could evolve into a scalable research tool or a commercial product that academic institutions, research organizations, or publishers could adopt to enhance their existing search systems. Additionally, the tool's potential to deliver more precise, context-aware search results and suggestions creates opportunities for commercialization in the academic publishing industry, where demand for enhanced digital research tools is on the rise. However, it is important to note that the full development of a commercial tool is not within the scope of this project.

5. Research Methodology

The research methodology consists of three key phases: system design and development, integration of the Computer Science Knowledge Graph (CS-KG), and evaluation and iteration. Each phase will focus on achieving the project's objectives efficiently while ensuring the successful integration of advanced AI techniques with the existing knowledge graph. The research will consist of the following steps:

- Knowledge Graph Utilization: Instead of traditional data collection, this project will utilize the pre-existing Computer Science Knowledge Graph (CS-KG), which contains 67 million statements about 24 million entities (e.g., tasks, methods, materials, and metrics) from 14.5 million scientific articles. The CS-KG focuses on content extracted from article abstracts and titles using a sophisticated pipeline, making it particularly useful for this project's goals of improving the contextual relevance of search results.
- Retrieval-Augmented Generation (RAG) Integration: A prototype system will be built. It combines RAG, an advanced retrieval and generative model, with the CS-KG. The goal is to develop a hybrid system that leverages the structured knowledge within CS-KG to retrieve and generate more accurate, contextually relevant research results. In this phase, both generative model architectures and various knowledge injection techniques will be implemented. This system will enable researchers to find publications based on abstracts and titles, using semantic relations within CS-KG (e.g., linking tasks, methods, and metrics) to provide meaningful suggestions beyond keyword-based search.
- Prototype Development: A web-based interface will be developed to enable users to input research queries (such as keywords, titles, or abstracts). The system will then retrieve and rank the most relevant publications, leveraging both the structured knowledge in CS-KG and the generative capabilities of RAG. The technology stack will include machine learning libraries such as PyTorch, semantic web libraries like RDFlib, and natural language processing (NLP) tools to ensure smooth integration with the CS-KG.
- Refining Semantic Relationships: The project will explore ways to further refine the semantic relationships extracted from CS-KG. This will involve analyzing how different research entities (e.g., tasks, methods, metrics) are interconnected to ensure that the retrieval system is capable of providing high-quality results that are contextual.
- Quantitative Evaluation: The system's performance will be evaluated using standard information retrieval metrics such as precision, recall, and F1-score. These metrics will help assess the system's ability to retrieve the most relevant research papers based on user queries. Comparisons will be made with traditional retrieval systems to demonstrate the improvements brought about by integrating CS-KG.
- User Studies (Qualitative Evaluation): A group of researchers and domain experts will test the system, providing qualitative feedback on its relevance. The goal is to measure the system's ability to improve research workflows, helping users quickly discover relevant, high-quality literature.
- Throughout the project, comprehensive documentation will be produced, detailing the technical challenges, solutions, and system architecture. This will provide a roadmap for future work in developing scholarly tools that combine AI models with knowledge graphs.

6. Literature Review and Critical Analysis of Related Work

Retrieving relevant research papers to substantiate scientific claims, provide evidence, or offer contrasting viewpoints has been an ongoing area of study in the scientific community. The roots of paper retrieval and recommendation research can be traced back to 2003 with the KDD Cup, organized by ACM SIGKDD, which involved predicting the number of citations a paper would receive over time [1].

In 2009, Tang et al. [2] developed an approach using Restricted Boltzmann Machines to model research papers based on their topics and used Kullback-Leibler divergence to identify similar papers suitable for referencing. Similarly, the method proposed in [3] emphasizes extracting local contexts around citations and offers tools to help researchers identify suitable references for a specific textual context. More recently, Cohan et al. [4] introduced an approach to assess the "citation worthiness" of scientific texts using a Multi-Layer Perceptron model and a dataset called "SciCite." In a related research stream, scholars have investigated citation intent—understanding the purpose or motivation behind citing specific sources. Ontologies like CiTO [5] provide detailed classifications of citation relationships, describing them through over 20 different types. Citation intent classification has also benefited from deep learning methods, as shown in [6], where Convolutional Neural Networks were used to semantically classify citation intents.

Apart from citation intent, other retrieval methods focus on identifying relevant research documents using user queries. Techniques like collaborative filtering and PageRank are applied, often leveraging seed papers to recommend similar work [7, 8]. In [9], the authors introduced a model that integrates embeddings from local contexts, aiming to match relevant papers to specific paragraphs. This model used neural networks with self-attention and additive attention mechanisms, which improved the ability to capture both structural and contextual relationships between papers. Deep learning advancements have significantly enhanced the retrieval of research papers by capturing their semantic and contextual information [10]. A recent study in 2024 [11] introduced ILCiteR, a tool that retrieves evidence spans from the scientific literature to help identify papers relevant to a given query, ranking both the evidence spans and associated papers for improved interpretability.

More recent efforts have explored the use of large language models (LLMs) for paper retrieval and citation generation tasks. For example, [12] evaluated LLMs such as Llama-2 and GPT-3.5 in generating citation text based on abstract and intent prompts. Another study [13] examined the trustworthiness of references suggested by LLMs in response to medical queries from the PubMedQA and MedQA datasets, noting discrepancies between LLM capabilities and expectations in providing reliable references. Additionally, a study in [14] explored the potential biases introduced by LLMs during the citation recommendation process, showing that LLMs could amplify existing biases in how scientific literature is structured and cited.

Different from prior work focusing on paper retrieval for citation recommendation, the proposed project emphasizes the integration of Knowledge Graph (KG) technologies with generative models to enhance scholarly paper retrieval for both research exploration and reference search. While previous efforts have largely focused on citation placement and intent analysis, this research project leverages the Computer Science Knowledge Graph (CS-KG) to semantically connect research papers through their abstracts and titles. By combining knowledge graphs with generative models, we aim to improve the accuracy of research paper retrieval, going beyond simple keyword-based searches to offer a deeper, more contextual understanding of the literature.

[1] J. Gehrke, P. Ginsparg, J. Kleinberg, Overview of the 2003 kdd cup, Acm Sigkdd Explorations Newsletter 5 (2) (2003) 149–151.
[2] J. Tang, J. Zhang, A discriminative approach to topic-based citation recommendation, in: Proceedings of the 13th Pacific-Asia Conference on Advances in Knowledge Discovery and Data Mining, PAKDD ’09, Springer-Verlag, Berlin, Heidelberg, 2009, p. 572–579.
[3] Q. He, J. Pei, D. Kifer, P. Mitra, L. Giles, Context-aware citation recommendation, in: Proceedings of the 19th international conference on World wide web, 2010, pp. 421–430.
[4] A. Cohan, W. Ammar, M. van Zuylen, F. Cady, Structural scaffolds for citation intent classification in scientific publications, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Association for Computational Linguistics, Minneapolis, Minnesota, 2019, pp. 3586–3596.

[5] S. Peroni, D. Shotton, Fabio and cito: ontologies for describing bibliographic resources and citations, Journal of Web Semantics 17 (2012) 33–43.
[6] A. Lauscher, G. Glavaˇs, S. P. Ponzetto, K. Eckert, Investigating convolutional networks and domain-specific embeddings for semantic classification of citations, in: Proceedings of the 6th International Workshop on Mining Scientific Publications, WOSP 2017, Association for Computing Machinery, New York, NY, USA, 2017, p. 24–28.
[7] C. Caragea, A. Silvescu, P. Mitra, C. L. Giles, Can't see the forest for the trees? a citation recommendation system, in: Proceedings of the 13th ACM/IEEE-CS joint conference on Digital libraries, 2013, pp. 111–114.
[8] H. Jia, E. Saule, An analysis of citation recommender systems: Beyond the obvious, in: Proceedings of the 2017 IEEE/ACM international conference on advances in social networks analysis and mining 2017, 2017, pp. 216–223.
[9] Y. Zhang, Q. Ma, Dual attention model for citation recommendation with analyses on explainability of attention mechanisms and qualitative experiments, Computational Linguistics 48 (2) (2022) 403–470.
[10] Z. Ali, P. Kefalas, K. Muhammad, B. Ali, M. Imran, Deep learning in citation recommendation models survey, Expert Systems with Applications 162 (2020) 113790.
[11] S. Ghosh Roy, J. Han, Ilciter: Evidence-grounded interpretable local citation recommendation, arXiv e-prints (2024) arXiv–2403.
[12] F. S¸ahinu¸c, I. Kuznetsov, Y. Hou, I. Gurevych, Systematic task exploration with llms: A study in citation text generation, arXiv preprint arXiv:2407.04046 (2024).
[13] K.Wu, E.Wu, A. Cassasola, A. Zhang, K.Wei, T. Nguyen, S. Riantawan, P. S. Riantawan, D. E. Ho, J. Zou, How well do llms cite relevant medical references? an evaluation framework and analyses, arXiv preprint arXiv:2402.02008 (2024).
[14] A. Algaba, C. Mazijn, V. Holst, F. Tori, S. Wenmackers, V. Ginis, Large language models reflect human citation patterns with a heightened citation bias, arXiv preprint arXiv:2405.15739 (2024).

7. Discuss any Related Ethical Issues

(Does the project contradict in any way the teachings of Islam or the Arabic Culture, Does the project involve using animals, humans, or samples from animals and humans, etc,…Does the research involve collecting information of special nature or implications, etc,…Do you need special approvals to obtain samples, or conduct any aspect of the research, Specify the kind of risks involved and what protection the PI has for the safety of subjects, etc,…)

The proposed project does not contradict the teachings of Islam or Arabic culture, as it focuses on improving scholarly tools through the use of knowledge graphs and artificial intelligence. It does not involve the use of animals, humans, or samples from either. No sensitive information of a special nature will be collected or used, and there are no risks associated with the research. Therefore, no special approvals are required for conducting any aspect of the project.

8. Cooperation with external institutions related to project (agreements, joint work, etc,…)

The project will be integrated into a long-term collaborative initiative involving the Principal Investigator (PI) and institutions, including the University of Cagliari (Italy) and the Knowledge Media Institute at The Open University (UK). Although this collaboration does not currently operate under a formal agreement, it has already yielded significant outcomes, contributing to the publication of 9 articles in Q1 journals and 14 conference and workshop papers over the past five years. This ongoing partnership enhances the project's foundation by bringing together diverse expertise and perspectives, fostering innovation in research methodologies, and enabling a more comprehensive approach to tackling complex challenges in the field. By leveraging these collaborative networks, the project aims to amplify its impact and advance knowledge in scholarly communication and knowledge graph technologies.

9. Tasks and milestones of the project (briefly describe each task and milestone)

Task 1. Data preparation and vectorization.
The first task involves comprehensive data preparation, which is critical for ensuring the integrity and relevance of the information to be used in the research project. More precisely, the existing knowledge graph formats (e.g., RDF and turtle) do not allow direct usage for the research project and transformation into suitable formats is necessary. Additionally, the vectorization of the knowledge graph information will also be performed to exploit semantic similarity retrieval approaches for the retrieval component of the proposed project.
M1. Successful transformation of the knowledge graph data into suitable formats and completion of the vectorization process for use in retrieval models.

Task 2. Retrieval model design.
Task 2 of the project includes the design of the retrieval model of the proposed research-based solution. For this, semantic search on the vector database, similarity search on the vector database, and semantic search on the knowledge graph, similarity search on the knowledge graph options will be explored. According to the current literature, there is no best strategy for this, and all the techniques need to be explored to find a balance between efficiency and computational time.

Task 3. Retrieval-Augmented Generation (RAG) Design.
Building upon the retrieval model, we will design a Retrieval-Augmented Generation (RAG) architecture that combines the strengths of both retrieval and generative AI models. This design will specify how the retrieval model retrieves relevant literature, which will then be used as input for a generative model, such as GPT or T5, to produce contextually rich and informative responses. By leveraging the structured knowledge from the CS-KG and the generative capabilities of these models, we aim to enhance the precision of search results significantly. Initial experiments will be conducted to evaluate the performance of the RAG setup and refine it based on the results to ensure optimal functionality
M2. Design and testing of the RAG architecture, with initial experiments conducted to evaluate and refine the performance of the model.

Task 4. Prototype Implementation.
The next step involves the implementation of a web-based prototype that embodies the integrated system developed in the previous tasks. This prototype will feature a user-friendly interface, allowing researchers to interact seamlessly with the retrieval and generative components. We will ensure that all system components are effectively integrated, facilitating smooth communication between the retrieval model, the generative AI, and the user interface. Before user studies, we will conduct internal testing to assess the functionality of the prototype, identifying and addressing any issues that may arise during this stage.
M3. Development and internal testing of the prototype, ensuring all components work seamlessly and identifying any issues for resolution before assessment.

Task 5. Evaluation and dissemination.
To assess the effectiveness of the developed prototype, the evaluation will be user-based to assess its effectiveness in aiding literature discovery. Researchers from various academic disciplines are expected to be invited to participate in this study, providing valuable feedback on their experiences while using the tool. We will collect both qualitative and quantitative data, including user satisfaction and search accuracy metrics. This comprehensive evaluation will enable us to analyze the prototype's performance, helping to identify areas for improvement and ensuring that the tool meets the needs of the research community. Ultimately, journals and conferences for publication and presentation will be targeted, ensuring that our work reaches a relevant audience. In preparation for these events, we will develop presentation materials, including slides and posters, to effectively communicate our research outcomes. Additionally, we will engage with the academic community through social media, webinars, and workshops, fostering

discussions and collaborations that can further enhance the impact of our work in the fields of semantic web technologies and machine learning.

M4. Completion of user studies, data collection, and analysis of the prototype's performance. Preparation and dissemination of research findings

10. Project Schedule of Execution of Research Project Tasks

Task	Year1												Year 2												Year 3										
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	11	12
Task-1	X	X	X																																
Task-2			X	X	X	X	X	X																											
Task-3					X	X	X	X	X	X	X																								
Task-4										X	X	X	X																						
Task-5											X	X	X	X	X																				

11. Budget of the Proposed Research

Budget must be in UAE Dirhams

No.	Item	Amount Required	Justifications (Explain why this item is essential for the project)
1	**Research Assistants**	10,000 30% of the overall budget	Funding for 1 research assistant
2	**Small Equipment**	800	External HardDisk (at least 2 TB size) for Backup purposes
3	**Testing, Analysis, and Field Work**	0	
4	**Consumables**	0	
5	**Publications**	4,000	Fees for publishing research papers in open-access journals. Open-access publishing ensures high visibility of the research outcome and the University of Sharjah.
6	**Travel (used for field test, data collection,…etc.)**	0	
7	**Conference Attendance (Only applied to competitive, targeted, and collaborative research proposals with at least two years duration)**	0	
8	**Software**	18,000	- 10,000 AED for experiments, storage, and big data handling in cloud-based services (e.g., to process the knowledge graph and run LLM functionality) - 8,000 AED for GPT API pricing to access state-of-the-art LLMs
9	**Stationary**	0	
10	**Other Items**	500	Miscellaneous expenses
	Total	32,500	

12. Available or Expected External Sources of Funds

The proposed seed project is limited in budget and there are no economic resources that can be added from external collaborators at the time of writing. However, thanks to the long-term collaborations, from the University of Cagliari, Italy, servers are made available for tasks related to the preparation of the information from CS-KG before starting computation on the cloud service.

13. Names and Contact Details Including Affiliation and Email of at least Five Potential Reviewers, avoiding conflict of interest.

- Dr. Dimitar Dimitrov, GESIS – Leibiniz Institute, Germany, (dimitar.dimitrov@gesis.org)
- Dr. Ivan Heibi, University of Bologna, Italy (ivan.heibi@unibo.it)
- Dr. Vivek Kumar, University of Federal Armed Forces, Germany (vivek.kumar@unibw.de)
- Dr. Leyla Castro, ZB Med, Germany (ljgarcia@zbmed.de)
- Dr. Mauro Dragoni, Bruno Klesser Foundation, Italy, (dragoni@fbk.eu)

Research Project Refereeing Criteria

	Criterion	
First	**Project Formulation and Project Objectives**	1- Is the problem well defined and indicates potential contribution to knowledge? 2- Do the objectives clearly address the problem and are achievable?
Second	**Expected Research Outputs**	Expected Publication of refereed books, book chapters, articles in refereed journals, conferences, or patents, software, products, creative works, etc,……..
Third	**Potential Project Impacts**	1- Is the project expected to add to scientific knowledge? 2- Is the project expected to contribute to the development of the community? 3- Possibility of innovation and potential commercial product.
Fourth	**Methodology**	1- Are the conceptual framework, design, methods, and analyses used in the project satisfactory and adequate? 2- Is the methodology appropriate to the objectives of the project in question? 3- Are the applicants aware of the potential problems likely to occur while conducting the research? Do they have alternative methods?
Fifth	**Researcher(s)**	1- Research track record of the investigator(s). 2- Integrated experience of the research team for this project.
Sixth	**Project Schedule**	Is the proposed schedule appropriate for achieving the project objectives?
Seventh	**Budget**	1- To what extent are the budget items commensurate with the project? 2- Is there an external source of fund for the project?

Office of Vice Chancellor For Research and Graduate Studies

Research Grant Processing Form

Project ID Number												

Project Title:		
Applicant's Name(Principal Investigator) **Danilo Dessì**	**Signature:**	**Date: 10.11.2024**

Chair of Academic College Research Committee.

The committee, in its meeting number……held on …….……, has considered the application and provided comments to the applicant to improve its quality and ensure its adherence to the bylaws.

Name:	**Signature:**	**Date:**

Dean of Academic College:

Name:	**Signature:**	**Date:**

Chair of Research Department

Received on: …./…../……..

Date of Sending the Project to Refereeing: …….…….

Date of Completion of Project Refereeing:…….…….

Refereeing Results:…….……..

Summary: ……

……..…….

……..…….

……..…….

……..…….

……..……